Gandaki University

Faculty of Science and Technology

Bachelor of Information Technology (BIT)

Project Progress Interaction Log Sheet

Academic Year: 2022 Semester: 8th Visit No: 1 Date of Visit:

Project Title: Dhoom Car Racing

Student Name(s): Aakash Rana Magar, Ayush Tamrakar, Roshan Thapa

Student ID(s):

Student–Supervisor Meeting Log (Minimum 5 interactions are required):

Completed Tasks/ Achievement	**Isssues/ challenges**	**Next Week plan**
1. Set up Unity project and installed packages: Photon PUN2, Firebase SDK, TextMesh Pro, SlimUI, Input System, Splines. 2. Imported 3D car models (Car_1 through Car_55) with LOD variants and interior models from FBX files. 3. Implemented basic car physics using Unity WheelColliders with 4-wheel setup (front-left, front-right, rear-left, rear-right) in PhotonCarController.cs. 4. Created main menu UI (MainMenuController.cs) with buttons for Single Player, Multiplayer, Garage, Profile, Google Sign In, and Quit. 5. Designed scene flow: MainMenu -> MultiplayerMenu -> Joining -> Lobby -> TrackSelection -> Race Tracks (11 scenes total). 6. Set up GameSession.cs singleton (DontDestroyOnLoad) to persist game state: game mode, selected car, selected track, room code, and lap count across scenes. 7. Implemented scene transitions with fade-in/fade-out animation using CanvasGroup alpha interpolation in SceneSwitcher.cs.	1. Difficulty tuning WheelCollider parameters (spring, damper, force) for realistic yet fun arcade-style driving feel. 2. DontDestroyOnLoad singletons (GameSession, AudioManager, FirebaseManager) persist across scenes causing initialization order issues. 3. Unity Input System and Legacy Input Manager coexist in the project causing occasional input conflicts.	1. Implement full engine simulation with RPM calculation, gear shifting (automatic + manual), torque curves, and rev limiter. 2. Add speed-dependent steering reduction and drift mechanics with counter-steer. 3. Create checkpoint system with sequential validation and lap tracking. 4. Build car selection (garage) UI with previous/next car cycling and 3D car preview.

Supervisor's Feedback:

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Supervisor Name & Signature: Er. Saroj Giri ________________

Date: _____________________